**SOK-3081 | Chapter 5 seminar**

Questions adapted from Hill, Griffiths and Lim, *Principles of Econometrics*, 5th edition. Use the matching worked-solutions document after attempting them.

## Exercise 5.1: Estimate a multiple regression by hand

Consider $y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + e_i$, with the following seven observations (textbook Table 5.5).

| $i$ | $y_i$ | $x_{i1}$ | $x_{i2}$ | $x_{i3}$ |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 1 |
| 2 | 1 | 1 | 1 | -2 |
| 3 | 4 | 1 | 2 | 2 |
| 4 | 0 | 1 | -2 | 1 |
| 5 | 1 | 1 | 1 | -2 |
| 6 | -2 | 1 | -2 | -1 |
| 7 | 2 | 1 | 0 | 1 |

Use a calculator or spreadsheet.

**(a)** Express the observations as deviations from their sample means: $x^*_{i2} = x_{i2} - \bar{x}_2$, $x^*_{i3} = x_{i3} - \bar{x}_3$, and $y^*_i = y_i - \bar{y}$.

**(b)** Calculate $\sum_i y^*_i x^*_{i2}$, $\sum_i (x^*_{i2})^2$, $\sum_i y^*_i x^*_{i3}$, $\sum_i x^*_{i2} x^*_{i3}$, and $\sum_i (x^*_{i3})^2$.

**(c)** Use the formulas in Appendix 5A to find $b_1$, $b_2$, and $b_3$.

**(d)** Find the seven least-squares residuals.

**(e)** Estimate the error variance $\sigma^2$.

**(f)** Find the sample correlation between $x_2$ and $x_3$.

**(g)** Find the standard error of $b_2$.

**(h)** Find SSE, SST, SSR, and $R^2$.

## Hints for Exercise 5.1

**(b)** These are the familiar cross-products and sums of squares. Use the short notation

$$\begin{aligned}
\sum_i y_i^* x_{i2}^* &= S_{y2}, \quad \sum_i (x_{i2}^*)^2 = S_{22}, \\
\sum_i y_i^* x_{i3}^* &= S_{y3}, \quad \sum_i x_{i2}^* x_{i3}^* = S_{23}, \\
\sum_i (x_{i3}^*)^2 &= S_{33}.
\end{aligned}$$

First calculate each product for every observation; then sum down each column.

**(c)** Define the common denominator

$$D = S_{22}S_{33} - S_{23}^2.$$

Then use

$$\begin{aligned}
b_2 &= \frac{S_{y2}S_{33} - S_{y3}S_{23}}{D}, \\
b_3 &= \frac{S_{y3}S_{22} - S_{y2}S_{23}}{D}, \\
b_1 &= \bar{y} - b_2\bar{x}_2 - b_3\bar{x}_3.
\end{aligned}$$

**(d)** For each observation, first calculate the fitted value and then the residual:

$$\hat{y}_i = b_1 + b_2 x_{i2} + b_3 x_{i3}, \qquad \hat{e}_i = y_i - \hat{y}_i.$$

**(e)** Square and add the residuals from part (d), then divide by the residual degrees of freedom:

$$SSE = \sum_{i=1}^{N} \hat{e}_i^2, \qquad \hat{\sigma}^2 = \frac{SSE}{N - K}.$$

Here $N$ is the number of observations and $K$ is the number of estimated coefficients, including the intercept.

**(f)** Use the quantities calculated in part (b):

$$r_{23} = \frac{S_{23}}{\sqrt{S_{22}S_{33}}}.$$

**(g)** First find the estimated variance of $b_2$, then take its square root:

$$\widehat{\mathrm{var}}(b_2 \mid \mathbf{X}) = \frac{\hat{\sigma}^2}{(1 - r_{23}^2)S_{22}}, \qquad se(b_2) = \sqrt{\widehat{\mathrm{var}}(b_2 \mid \mathbf{X})}.$$

**(h)** Use

$$
\begin{aligned}
SSE &= \sum_i \hat{e}_i^2, \\
SST &= \sum_i (y_i - \bar{y})^2, \\
SSR &= SST - SSE, \\
R^2 &= \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.
\end{aligned}
$$

## Exercise 5.2: Confidence interval and hypothesis test

Use your results from Exercise 5.1.

**(a)** Compute a 95% confidence interval for $\beta_2$.

**(b)** Test $H_0 : \beta_2 = 1.25$ against $H_1 : \beta_2 \neq 1.25$. State the significance level you use and your conclusion.

## Hints for Exercise 5.2

**(a)** Use the estimate and standard error from Exercise 5.1:

$$b_2 \pm t_{0.975,\,N-K}\, se(b_2).$$

There are $N - K$ residual degrees of freedom. Find the corresponding critical value from the $t$ distribution before calculating the lower and upper endpoints.

**(b)** This is a two-sided test. The test statistic is

$$t = \frac{b_2 - 1.25}{se(b_2)}.$$

At the 5% significance level, compare $|t|$ with $t_{0.975,\,N-K}$. Reject $H_0$ if $|t|$ is larger than the critical value. Equivalently, calculate the two-sided p-value $2P(T_{N-K} > |t|)$ and compare it with 0.05. As a final check, ask whether 1.25 lies inside the 95% confidence interval from part (a).

### Exercise 5.3: Complete and interpret regression output

For 1200 London households, estimate $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + e$. $WALC$ is the percentage of the budget spent on alcohol; $TOTEXP$ is total expenditure, $NK$ the number of children, and $AGE$ the household head's age. The supplied output (Table 5.6) is incomplete.

| Variable | Coefficient | Standard error | $t$ statistic | p-value |
|---|---|---|---|---|
| Intercept | 1.4515 | 2.2019 | ? | 0.5099 |
| $\ln(TOTEXP)$ | 2.7648 | ? | 5.7103 | 0.0000 |
| $NK$ | ? | 0.3695 | -3.9376 | 0.0001 |
| $AGE$ | -0.1503 | 0.0235 | -6.4019 | 0.0000 |

Additional output: $SSE = 46221.62$, sample mean of $WALC = 6.19434$, and sample standard deviation of $WALC = 6.39547$.

**(a)** Find the missing intercept $t$ statistic, standard error of $b_2$, estimate $b_3$, $R^2$, and regression standard error $\hat{\sigma}$.

**(b)** Interpret $b_2$, $b_3$, and $b_4$.

**(c)** Compute a 95% confidence interval for $\beta_4$. Interpret it.

**(d)** Which coefficients differ significantly from zero at 5%? Explain.

**(e)** At 5%, test whether an additional child reduces the mean alcohol budget share by 2 percentage points against a two-sided alternative.

## Hints for Exercise 5.3

There are $N = 1200$ observations and $K = 4$ estimated coefficients, including the intercept. Thus, use $N - K$ residual degrees of freedom.

**(a)** Complete the regression output by rearranging the usual t-statistic:

$$t_k = \frac{b_k}{se(b_k)}, \qquad se(b_k) = \frac{b_k}{t_k}, \qquad b_k = t_k\, se(b_k).$$

The reported sample standard deviation uses $N - 1$ in its denominator. For $R^2$ and the regression standard error, use

$$SST = (N-1)s_y^2, \qquad R^2 = 1 - \frac{SSE}{SST}, \qquad \hat{\sigma} = \sqrt{\frac{SSE}{N-K}}.$$

**(b)** Interpret every coefficient while holding the other included variables constant. Since $\ln(TOTEXP)$ is used, a 1% increase in $TOTEXP$ changes mean $WALC$ by approximately $b_2/100$ **percentage points**. For $NK$ and $AGE$, interpret a one-unit increase directly. Pay attention to the signs and distinguish percentage points from percent.

**(c)** Use

$$b_4 \pm t_{0.975,\, N-K}\, se(b_4).$$

Interpret both endpoints as changes in the mean alcohol budget share, in percentage points per additional year of age.

**(d)** Each reported p-value tests $H_0 : \beta_k = 0$ against $H_1 : \beta_k \neq 0$. At the 5% level, reject $H_0$ when the p-value is at most 0.05. Equivalently, compare $|t_k|$ with the two-sided critical value.

**(e)** "Reduces by 2 percentage points" corresponds to $H_0 : \beta_3 = -2$. Use the two-sided statistic

$$t = \frac{b_3 - (-2)}{se(b_3)} = \frac{b_3 + 2}{se(b_3)}.$$

Compare $|t|$ with $t_{0.975,\, N-K}$, or compare the two-sided p-value with 0.05. Remember that testing $\beta_3 = -2$ is not the same as testing $\beta_3 = 0$.

## Exercise 5.4: Alcohol expenditure, children and age

For 1200 households, the model is $WALC = \beta_1 + \beta_2 \ln(TOTEXP) + \beta_3 NK + \beta_4 AGE + \beta_5 AGE^2 + e$. Definitions are as in Exercise 5.3. Table 5.7 supplies $\widehat{WALC} = 8.149 + 2.884 \ln(TOTEXP) - 1.217NK - 0.5699AGE + 0.005515AGE^2$.

The relevant estimated covariance matrix is:

| | $b_3$ $(NK)$ | $b_4$ $(AGE)$ | $b_5$ $(AGE^2)$ |
|---|---|---|---|
| $b_3$ | 0.1462 | -0.01774 | 0.0002347 |
| $b_4$ | -0.01774 | 0.03204 | -0.0004138 |
| $b_5$ | 0.0002347 | -0.0004138 | 0.000005438 |

**(a)** Find a point estimate and 95% confidence interval for the change in mean alcohol budget share from one additional child.

**(b)** Find point estimates and 95% confidence intervals for the marginal effect of age at ages (i) 25, (ii) 50, and (iii) 75.

**(c)** Find a point estimate and 95% confidence interval for the age at which the mean alcohol budget share is minimized.

**(d)** Summarize what (a)-(c) tell you.

**(e)** If errors conditional on $\mathbf{X}$ are normal, which intervals are valid in finite samples, and which require a large-sample approximation?

**(f)** Answer (e) if the conditional errors are not normal.

## Hints for Exercise 5.4

There are $N = 1200$ observations and $K = 5$ estimated coefficients, including the intercept. Use $N - K$ residual degrees of freedom for the t critical value.

**(a)** The effect of one additional child is $\beta_3$. Read its point estimate from the fitted equation. Its variance is the diagonal covariance matrix entry for $b_3$:

$$se(b_3) = \sqrt{\widehat{\text{var}}(b_3)}, \qquad b_3 \pm t_{0.975,\, N-K} se(b_3).$$

**(b)** Because both $AGE$ and $AGE^2$ are included, the marginal effect at age $a$ is

$$m(a) = \frac{\partial E(WALC \mid \mathbf{X})}{\partial AGE} = \beta_4 + 2a\beta_5.$$

Estimate it using $\widehat{m}(a) = b_4 + 2ab_5$. For its standard error, use the variance formula for a linear combination:

$$\widehat{\text{var}}[\widehat{m}(a)] = \widehat{\text{var}}(b_4) + 4a^2\widehat{\text{var}}(b_5) + 4a\widehat{\text{cov}}(b_4, b_5).$$

Then calculate

$$\widehat{m}(a) \pm t_{0.975,\, N-K} \sqrt{\widehat{\text{var}}[\widehat{m}(a)]}$$

separately for $a = 25$, 50, and 75.

**(c)** At the minimum, the marginal effect of age equals zero. Solve $\beta_4 + 2a^*\beta_5 = 0$:

$$\hat{a}^* = -\frac{b_4}{2b_5}.$$

Check that $b_5 > 0$, so the turning point is a minimum. For an approximate standard error, let $g(b_4, b_5) = -b_4/(2b_5)$ and use the delta method with

$$g_4 = -\frac{1}{2b_5}, \qquad g_5 = \frac{b_4}{2b_5^2},$$

and

$$\widehat{\text{var}}(\hat{a}^*) \approx g_4^2\widehat{\text{var}}(b_4) + g_5^2\widehat{\text{var}}(b_5) + 2g_4g_5\widehat{\text{cov}}(b_4, b_5).$$

Take the square root and form the approximate 95% interval in the usual way.

**(d)** Summarize direction, magnitude, and uncertainty. Ask whether each interval contains zero. Describe the fitted age pattern before and after the turning point, but avoid making a causal claim without further assumptions.

**(e)** Under conditionally normal errors and MR1–MR6, intervals for linear combinations of coefficients have exact finite-sample t justification. The turning point is a nonlinear ratio of coefficients, so its delta-method interval remains a large-sample approximation.

**(f)** Without conditional normality, the usual finite-sample t results are not exact. Under the other assumptions and suitable large-sample conditions, all three interval procedures can instead be justified approximately.

### Exercise 5.5: Does additional data ensure consistency?

Assume MR1-MR6. For each model, find $\text{var}(b_2 \mid \mathbf{x})$ and examine consistency by checking whether this variance tends to zero as $T \to \infty$.

**(a)** $y_t = \beta_1 + \beta_2 t + e_t$, for $t = 1, \ldots, T$.

Useful identities: $\sum_{t=1}^{T} t = T(T+1)/2$, $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$, and $\sum_t (t - \bar{t})^2 = \sum_t t^2 - (\sum_t t)^2/T$.

**(b)** $y_t = \beta_1 + \beta_2 (0.5)^t + e_t$, for $t = 1, \ldots, T$.

Useful identity: $r + r^2 + \cdots + r^n = r(1 - r^n)/(1 - r)$.

**(c)** Give an intuitive explanation for your results.

## Hints for Exercise 5.5

Both models contain an intercept and one explanatory variable. Begin with

$$\operatorname{var}(b_2 \mid \mathbf{x}) = \frac{\sigma^2}{S_{xx}}, \qquad S_{xx} = \sum_{t=1}^{T}(x_t - \bar{x})^2 = \sum_t x_t^2 - \frac{(\sum_t x_t)^2}{T}.$$

Consistency requires the estimation uncertainty to disappear as the sample grows. Examine whether $S_{xx} \to \infty$, or equivalently whether the slope variance tends to zero.

**(a)** Set $x_t = t$ and substitute the two supplied summation formulas into $S_{xx}$. Simplify before taking the limit. It can help to factor out $T(T+1)$ and compare the highest powers of $T$ in the numerator and denominator of the variance.

**(b)** Here $x_t = 2^{-t}$ and $x_t^2 = 4^{-t}$. Apply the geometric-series formula separately to $\sum_t x_t$ and $\sum_t x_t^2$, then insert both sums into $S_{xx}$. Take the limit of each term. Ask whether the later observations, whose $x_t$ values are very close to zero, keep adding meaningful variation.

**(c)** Compare what happens to the **range and accumulated variation** of the regressor. More observations help estimate a slope only when they add useful variation in $x$, not merely when they increase the sample count.

### Exercise 5.6: Test linear combinations

For a regression with three coefficients and $N = 63$, the estimates are $b_1 = 2$, $b_2 = 3$, and $b_3 = -1$. Their estimated covariance matrix is:

| | $b_1$ | $b_2$ | $b_3$ |
|---|---|---|---|
| $b_1$ | 3 | -2 | 1 |
| $b_2$ | -2 | 4 | 0 |
| $b_3$ | 1 | 0 | 3 |

Use a 5% significance level and a two-sided alternative to test:

**(a)** $H_0 : \beta_2 = 0$.

**(b)** $H_0 : \beta_1 + 2\beta_2 = 5$.

**(c)** $H_0 : \beta_1 - \beta_2 + \beta_3 = 4$.

## Hints for Exercise 5.6

There are $N - K = 63 - 3$ residual degrees of freedom. Every alternative is two-sided, so compare $|t|$ with $t_{0.975,\, N-K}$.

For a linear combination $L = \mathbf{c}'\beta$, use

$$\widehat{L} = \mathbf{c}'\mathbf{b}, \qquad \widehat{\operatorname{var}}(\widehat{L}) = \mathbf{c}'\widehat{\operatorname{Var}}(\mathbf{b})\mathbf{c}, \qquad t = \frac{\widehat{L} - L_0}{se(\widehat{L})}.$$

When calculating by hand, include every variance and covariance term:

$$\widehat{\operatorname{var}}\left(\sum_j c_j b_j\right) = \sum_j c_j^2 \widehat{\operatorname{var}}(b_j) + 2\sum_{j<k} c_j c_k \widehat{\operatorname{cov}}(b_j, b_k).$$

**(a)** Use $\mathbf{c} = (0, 1, 0)'$ and $L_0 = 0$. The standard error of a single coefficient is the square root of its diagonal covariance-matrix entry.

**(b)** Use $\mathbf{c} = (1, 2, 0)'$ and $L_0 = 5$. Pay attention to the covariance between $b_1$ and $b_2$.

**(c)** Use $\mathbf{c} = (1, -1, 1)'$ and $L_0 = 4$. The negative weight on $b_2$ changes the signs of covariance contributions involving that coefficient.

### Exercise 5.7: Precision, correlation and tests

For $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + e$, $N = 203$. You are given $\sum_i (x_{i2} - \bar{x}_2)^2 = 1780.7$, $\sum_i (x_{i3} - \bar{x}_3)^2 = 3453.3$, $b_2 = 0.7176$, $b_3 = 1.0516$, $SSE = 6800.0$, and $r_{23} = 0.7087$.

**(a)** Find the standard errors of $b_2$ and $b_3$.

**(b)** At 5%, test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$.

**(c)** At 10%, test $H_0 : \beta_3 \leq 0.9$ against $H_1 : \beta_3 > 0.9$.

**(d)** Given $\widehat{\text{cov}}(b_2, b_3) = -0.019521$, test $H_0 : \beta_2 = \beta_3$ against $H_1 : \beta_2 \neq \beta_3$ at 1%.

## Hints for Exercise 5.7

There are $N - K = 203 - 3$ residual degrees of freedom. First estimate the error variance:

$$\hat{\sigma}^2 = \frac{SSE}{N - K}.$$

**(a)** With two explanatory variables, use

$$se(b_2) = \sqrt{\frac{\hat{\sigma}^2}{S_{22}(1 - r_{23}^2)}}, \qquad se(b_3) = \sqrt{\frac{\hat{\sigma}^2}{S_{33}(1 - r_{23}^2)}}.$$

Here $S_{22}$ and $S_{33}$ are the two supplied centered sums of squares. Notice how correlation between the regressors affects both standard errors.

**(b)** Use the two-sided statistic $t = (b_2 - 0)/se(b_2)$ and compare $|t|$ with $t_{0.975,\, N-K}$.

**(c)** This is an upper-tail test. Calculate the statistic at the null boundary, $t = (b_3 - 0.9)/se(b_3)$, and compare it with $t_{0.90,\, N-K}$. The point estimate being above 0.9 is not by itself enough to reject the null.

**(d)** Rewrite the null as $H_0 : \beta_2 - \beta_3 = 0$. Use

$$\widehat{\text{var}}(b_2 - b_3) = se(b_2)^2 + se(b_3)^2 - 2\widehat{\text{cov}}(b_2, b_3).$$

Then form the t-statistic for the estimated difference. At 1% with a two-sided alternative, use $t_{0.995,\, N-K}$.

## Exercise 5.8: Medals at the 1996 Olympics

For the 79 countries winning at least one medal, $MEDALS$ is the medal count, $POPM$ population in millions, and $GDPB$ GDP in billions of 1995 dollars. Assume MR1-MR6. The fitted model has $R^2 = 0.4879$: $\widehat{MEDALS} = 5.917 + 0.01813POPM + 0.01026GDPB$. Standard errors, in coefficient order: (1.510, 0.00819, 0.00136).

**(a)** Interpret the population and GDP coefficient estimates.

**(b)** Interpret $R^2$.

**(c)** Test no relationship with GDP against a positive relationship at 1%. Does the conclusion change at 5%?

**(d)** Repeat (c) for population.

**(e)** At 5%, test $H_0 : \beta_2 = c$ against $H_1 : \beta_2 \neq c$ for (i) $c = 0.01$, (ii) 0.02, (iii) 0.03, and (iv) 0.04. Are the results contradictory? Explain.

**(f)** Find a 95% confidence interval for $\beta_2$ and comment on it.

## Hints for Exercise 5.8

The model has $N - K = 79 - 3$ residual degrees of freedom. Keep track of the units: population is measured in millions and GDP in billions of dollars.

**(a)** Interpret each slope while holding the other explanatory variable constant. A conditional mean medal count need not be an integer.

**(b)** $R^2$ is the proportion of the sample variation in medal counts around their mean explained by the two regressors. It is not the percentage of medals predicted correctly and does not establish causality.

**(c)** For GDP, use the upper-tail statistic

$$t = \frac{b_3 - 0}{se(b_3)}.$$

Compare it first with $t_{0.99,\,N-K}$ and then with $t_{0.95,\,N-K}$. A lower significance level requires stronger evidence.

**(d)** Repeat the same upper-tail procedure using $b_2$ and $se(b_2)$ for population.

**(e)** For each proposed value $c$, calculate

$$t = \frac{b_2 - c}{se(b_2)}$$

and use the two-sided 5% critical value. Several different null values can all fail to be rejected: failure to reject is not proof that a null is true.

**(f)** Form

$$b_2 \pm t_{0.975,\,N-K} se(b_2).$$

Compare the four values in part (e) with the interval. A two-sided 5% test rejects exactly those null values that lie outside the corresponding 95% confidence interval.

## Exercise 5.9: Medals at the 1992 Olympics

Among the 64 medal-winning countries, exclude the United Kingdom and estimate with $N = 63$. Definitions are as in Exercise 5.8. Assume MR1-MR6. The fitted model has $R^2 = 0.275$: $\widehat{MEDALS} = -13.153 + 2.764\ln(POPM) + 4.270\ln(GDPB)$. Standard errors: (5.974, 2.070, 1.718).

**(a)** Interpret the population and GDP coefficients.

**(b)** Interpret $R^2$.

**(c)** Test no relationship with GDP against a positive relationship at 10%. What happens at 5%?

**(d)** Repeat (c) for population.

**(e)** Find point and 95% interval estimates for the UK's expected medal count, with $POPM = 58$ and $GDPB = 1010$. The standard error of $b_1 + \ln(58)b_2 + \ln(1010)b_3$ is 4.22196.

**(f)** The UK won 20 medals. Is the model useful for predicting its mean medal count? Find an approximate two-sided p-value for the null that its expected count equals 20.

**(g)** Write the expression for the standard error in (e), without evaluating it.

## Hints for Exercise 5.9

The regression uses $N - K = 63 - 3$ residual degrees of freedom. This is a **level-log model**: the dependent variable is in levels and both explanatory variables enter as natural logarithms.

**(a)** Holding the other regressor constant, a 1% increase in $X$ changes expected medals by approximately $b/100$ medal units. For a finite $p$% increase, the exact fitted change is $b \ln(1 + p/100)$.

**(b)** Interpret $R^2$ as the explained proportion of the sample variation in medal counts. Do not compare it mechanically with Exercise 5.8 because both the year and sample differ.

**(c)** For GDP, use $t = b_3/se(b_3)$ in an upper-tail test. Compare it with $t_{0.90,\,N-K}$ and $t_{0.95,\,N-K}$.

**(d)** Repeat the upper-tail test using the population coefficient and its standard error.

**(e)** Substitute the UK's values into

$$\widehat{MEDALS}_0 = b_1 + b_2 \ln(58) + b_3 \ln(1010).$$

The standard error of this estimated conditional mean is supplied. Construct the 95% interval using $t_{0.975,\,N-K}$. This is a confidence interval for the **mean**, not a prediction interval for one realized medal count.

**(f)** Let $\widehat{L}$ denote the estimated mean from (e). Test $H_0 : L = 20$ with

$$t = \frac{\widehat{L} - 20}{se(\widehat{L})},$$

and calculate a two-sided p-value. Distinguish statistical compatibility with 20 from an assessment of the model's overall predictive quality.

**(g)** Let $a = \ln(58)$ and $c = \ln(1010)$. For $\widehat{L} = b_1 + ab_2 + cb_3$, write

$$\begin{aligned}\widehat{\mathrm{var}}(\widehat{L}) = \widehat{\mathrm{var}}(b_1) + a^2\widehat{\mathrm{var}}(b_2) + c^2\widehat{\mathrm{var}}(b_3) \\ + 2a\widehat{\mathrm{cov}}(b_1, b_2) + 2c\widehat{\mathrm{cov}}(b_1, b_3) + 2ac\widehat{\mathrm{cov}}(b_2, b_3).\end{aligned}$$

The standard error is the square root. The three coefficient standard errors alone are insufficient because the covariance terms also matter.

## Exercise 5.10: Wheat yield and rainfall

For Mullewa, Western Australia, $T = 47$ annual observations (1950-1996) give $\widehat{YIELD}_t = 0.1717 + 0.01117t + 0.05238RAIN_t$. Standard errors: (0.1537, 0.00262, 0.01367). Yield is tonnes per hectare, $t = 1, \ldots, 47$, and rainfall is growing-season rainfall in inches. Its sample mean is 10.059 and standard deviation 2.624.

**(a)** Assuming MR1-MR5, interpret the trend and rainfall coefficients.

**(b)** At 5%, test whether technological change increases mean annual yield by no more than 0.01 tonnes per hectare against a larger increase.

**(c)** At 5%, test whether an extra inch of rain increases mean yield by 0.03 tonnes per hectare against a two-sided alternative.

**(d)** Adding squared rainfall gives $\widehat{YIELD}_t = -0.6759 + 0.011671t + 0.2229RAIN_t - 0.008155RAIN_t^2$. Standard errors: (0.3875, 0.00250, 0.0734, 0.003453). Why include squared rainfall? Does its coefficient have the expected sign?

**(e)** Repeat (b) using the quadratic model.

**(f)** Repeat (c) at mean rainfall using the quadratic model. For its rainfall coefficients, $\widehat{\text{cov}}(b_3, b_4) = -0.0002493$.

**(g)** Forecast yield in 1997 when rainfall was 9.48 inches, using (d).

**(h)** Before observing 1997 rainfall, what forecast would you make using each model? State your assumptions about rainfall.

## Hints for Exercise 5.10

The linear model has $T - K = 47 - 3$ residual degrees of freedom; the quadratic model has $47 - 4$. Since MR6 is not assumed, treat the usual t procedures as large-sample approximations.

**(a)** Interpret the trend coefficient as the change in mean yield associated with one additional year, holding rainfall fixed. Interpret the rainfall coefficient as the change associated with one additional inch, holding the time trend fixed. Keep the yield unit, tonnes per hectare, in both statements.

**(b)** "No more than 0.01" belongs in the null. Test at its boundary:

$$t = \frac{b_2 - 0.01}{se(b_2)}.$$

This is an upper-tail test, so compare with $t_{0.95,\,T-K}$.

**(c)** Use the two-sided statistic

$$t = \frac{b_3 - 0.03}{se(b_3)}$$

and compare $|t|$ with $t_{0.975,\,T-K}$.

**(d)** A quadratic term allows the rainfall slope to change with rainfall:

$$\frac{\partial E(YIELD \mid \mathbf{X})}{\partial RAIN} = \beta_3 + 2\beta_4 RAIN.$$

Think about crop yield when conditions move from dry to well watered and then to excessively wet. Use that story to assess the expected sign of $\beta_4$.

**(e)** Repeat part (b) with the trend estimate and standard error from the quadratic model. Remember that adding a coefficient changes the residual degrees of freedom.

**(f)** At rainfall level $r$, estimate the marginal effect by $\widehat{m}(r) = b_3 + 2rb_4$. Its estimated variance is

$$\widehat{\text{var}}[\widehat{m}(r)] = se(b_3)^2 + 4r^2 se(b_4)^2 + 4r\widehat{\text{cov}}(b_3, b_4).$$

Use $r = 10.059$, take the square root, and test $H_0 : m(r) = 0.03$ with a two-sided t-statistic. Do not omit the covariance term.

**(g)** Since 1996 corresponds to $t = 47$, use $t = 48$ for 1997. Insert the observed rainfall value into every rainfall term in the quadratic fitted equation.

**(h)** Before rainfall is observed, state a model for its uncertainty. A simple choice is to use the historical mean and variance. In the linear model, use $E(RAIN)$. In the quadratic model, remember

$$E(RAIN^2) = \text{var}(RAIN) + [E(RAIN)]^2.$$

Substituting $[E(RAIN)]^2$ alone gives fitted yield at mean rainfall, not the average fitted yield over uncertain rainfall.

## Exercise 5.11: A life-cycle wage profile

Consider $WAGE = \beta_1 + \beta_2 EDUC + \beta_3 EXPER + \beta_4 EXPER^2 + e$. Education and experience are measured in years.

**(a)** Find the marginal effect of experience on mean wage.

**(b)** What signs do you expect for $\beta_2$, $\beta_3$, and $\beta_4$? Explain.

**(c)** Express in terms of the parameters the experience level after which mean wage begins to decline.

**(d)** Using 600 observations, the fitted equation is $\widehat{WAGE} = -16.308 + 2.329 EDUC + 0.5240 EXPER - 0.007582 EXPER^2$. Standard errors: (2.745, 0.163, 0.1263, 0.002532). The estimated covariance of the experience coefficients is $\widehat{\text{cov}}(b_3, b_4) = -0.00030526$. Find 95% confidence intervals for:

- **(i)** The marginal effect of education on mean wage.
- **(ii)** The marginal effect of experience at $EXPER = 4$.
- **(iii)** The marginal effect of experience at $EXPER = 25$.
- **(iv)** The experience level after which mean wage declines.

## Hints for Exercise 5.11

There are $N - K = 600 - 4$ residual degrees of freedom in part (d).

**(a)** Differentiate the conditional mean with respect to $EXPER$. Both the linear and squared experience terms contribute:

$$\frac{\partial E(WAGE \mid \mathbf{X})}{\partial EXPER} = \beta_3 + 2\beta_4 EXPER.$$

**(b)** Use human-capital reasoning. Education and early experience are expected to raise wages, while diminishing returns to experience suggest a concave life-cycle wage profile.

**(c)** Set the marginal effect from part (a) equal to zero and solve for $EXPER^*$. Check the sign of $\beta_4$ to determine whether the turning point is a maximum or minimum.

**(d)(i)** The education marginal effect is $\beta_2$, so use

$$b_2 \pm t_{0.975,\, N-K}\, se(b_2).$$

**(d)(ii)–(iii)** At experience $a$, use

$$\widehat{m}(a) = b_3 + 2ab_4$$

and

$$\widehat{\mathrm{var}}[\widehat{m}(a)] = se(b_3)^2 + 4a^2 se(b_4)^2 + 4a\widehat{\mathrm{cov}}(b_3, b_4).$$

Evaluate separately at $a = 4$ and $a = 25$, then form the two 95% intervals.

**(d)(iv)** The turning point is the nonlinear function $g(b_3, b_4) = -b_3/(2b_4)$. For a delta-method standard error, use

$$g_3 = -\frac{1}{2b_4}, \qquad g_4 = \frac{b_3}{2b_4^2},$$

and

$$\widehat{\mathrm{var}}[g(b_3, b_4)] \approx g_3^2 \widehat{\mathrm{var}}(b_3) + g_4^2 \widehat{\mathrm{var}}(b_4) + 2g_3 g_4 \widehat{\mathrm{cov}}(b_3, b_4).$$

The resulting interval is approximate because it concerns a ratio of estimated coefficients.

## Exercise 5.12: House prices and residual regressions

For 850 houses sold in Baton Rouge in mid-2005, $PRICE$ is in thousands of dollars, $SQFT$ in hundreds of square feet, and $AGE$ in years. Regress $PRICE = \alpha_1 + \alpha_2 AGE + v$ and $SQFT = \delta_1 + \delta_2 AGE + u$. The fitted residuals satisfy $\sum_i \hat{v}_i^2 = 10377817$, $\sum_i \hat{u}_i^2 = 75773.4$, and $\sum_i \hat{u}_i \hat{v}_i = 688318$.

**(a)** Find the OLS estimate of $\beta_2$ in $PRICE = \beta_1 + \beta_2 SQFT + \beta_3 AGE + e$.

**(b)** Let $\hat{e}_i = \hat{v}_i - b_2 \hat{u}_i$. Show that $\sum_i \hat{e}_i^2 = \sum_i \hat{v}_i^2 - b_2 \sum_i \hat{v}_i \hat{u}_i$.

**(c)** Estimate the error variance $\sigma^2$.

**(d)** Find the standard error of $b_2$.

**(e)** Find an approximate p-value for $H_0 : \beta_2 \geq 9.5$ against $H_1 : \beta_2 < 9.5$. What do you conclude?

## Hints for Exercise 5.12

The two preliminary regressions remove the linear effect of $AGE$ from both $PRICE$ and $SQFT$. Their residuals are the parts of the variables not explained by house age.

**(a)** By the Frisch–Waugh–Lovell theorem, the size slope in the full multiple regression equals the slope from regressing $\hat{v}_i$ on $\hat{u}_i$:

$$b_2 = \frac{\sum_i \hat{u}_i \hat{v}_i}{\sum_i \hat{u}_i^2}.$$

Interpret the result using the stated units for price and square feet.

**(b)** Start from $\hat{e}_i = \hat{v}_i - b_2 \hat{u}_i$, square both sides, and sum over observations. Then use the identity from part (a), $b_2 \sum_i \hat{u}_i^2 = \sum_i \hat{u}_i \hat{v}_i$, to simplify the final term.

**(c)** Use the identity established in (b) to find the full-model $SSE$. The full regression estimates three coefficients, so

$$\hat{\sigma}^2 = \frac{SSE}{850 - 3}.$$

**(d)** The relevant remaining variation in size is $\sum_i \hat{u}_i^2$. Therefore,

$$se(b_2) = \sqrt{\frac{\hat{\sigma}^2}{\sum_i \hat{u}_i^2}}.$$

Use the full model's residual degrees of freedom.

**(e)** This is a lower-tail test. At the null boundary, calculate

$$t = \frac{b_2 - 9.5}{se(b_2)}.$$

The approximate p-value is $P(T_{847} \leq t)$. Retain unrounded values because the result may be close to the 5% threshold.

## Exercise 5.13: Okun's law

Okun's law is $U_t - U_{t-1} = -\gamma(G_t - G_N)$, where $U_t$ is the unemployment rate, $G_t$ economic growth, $G_N$ the growth rate needed to keep unemployment constant, and $0 < \gamma < 1$.

**(a)** Write the model as $DU_t = \beta_1 + \beta_2 G_t$, defining $DU_t = U_t - U_{t-1}$, $\beta_1 = \gamma G_N$, and $\beta_2 = -\gamma$.

**(b)** Quarterly US data, 1970 Q1-2014 Q4, give $\widehat{DU}_t = 0.1989 - 0.2713 G_t$, with residual standard deviation $\hat{\sigma} = 0.2749$. Find estimates of $\gamma$ and $G_N$. The estimated coefficient covariance matrix is:

| | $b_1$ | $b_2$ |
|---|---|---|
| $b_1$ | 0.0007212 | -0.0004277 |
| $b_2$ | -0.0004277 | 0.0006113 |

**(c)** Find standard errors of $b_1$, $b_2$, $\hat{\gamma}$, and $\widehat{G}_N$. Does each differ significantly from zero at 5%?

**(d)** At 5%, test whether $G_N = 0.8\%$ per quarter against a two-sided alternative.

**(e)** Find a 95% confidence interval for the adjustment coefficient.

**(f)** Find a 95% confidence interval for $E(U_{2015Q1} \mid U_{2014Q4} = 5.7991,\ G_{2015Q1} = 0.062)$.

**(g)** Find the corresponding 95% prediction interval for $U_{2015Q1}$. Explain how it differs from (f).

## Hints for Exercise 5.13

There are 180 quarterly observations and two regression coefficients, so use $180 - 2$ residual degrees of freedom.

**(a)** Expand the right-hand side:

$$-\gamma(G_t - G_N) = \gamma G_N - \gamma G_t.$$

Match the constant and slope with $\beta_1 + \beta_2 G_t$.

**(b)** From the parameter definitions,

$$\hat{\gamma} = -b_2, \qquad \widehat{G_N} = -\frac{b_1}{b_2}.$$

Keep growth in the units used in the fitted equation.

**(c)** Obtain $se(b_1)$ and $se(b_2)$ by taking square roots of the diagonal covariance-matrix entries. Since $\hat{\gamma} = -b_2$, $se(\hat{\gamma}) = se(b_2)$. For $g(b_1, b_2) = -b_1/b_2$, use the delta-method derivatives

$$g_1 = -\frac{1}{b_2}, \qquad g_2 = \frac{b_1}{b_2^2},$$

and calculate $g_1^2 \widehat{\text{var}}(b_1) + g_2^2 \widehat{\text{var}}(b_2) + 2 g_1 g_2 \widehat{\text{cov}}(b_1, b_2)$. Divide each estimate by its standard error to test against zero.

**(d)** You may test the nonlinear ratio using the delta-method standard error from (c). A useful alternative is the equivalent linear restriction

$$H_0 : \beta_1 + 0.8\beta_2 = 0.$$

For this route, use the variance formula for a linear combination and a two-sided 5% test.

**(e)** Form $\hat{\gamma} \pm t_{0.975,\, N-K} se(\hat{\gamma})$.

**(f)** First predict the unemployment change and add the known previous rate:

$$\widehat{U}_0 = U_{2014Q4} + b_1 + b_2 G_{2015Q1}.$$

Because the previous unemployment rate is treated as known, the estimated conditional-mean variance is

$$\widehat{\text{var}}(b_1 + b_2 G_0) = \widehat{\text{var}}(b_1) + G_0^2 \widehat{\text{var}}(b_2) + 2 G_0 \widehat{\text{cov}}(b_1, b_2).$$

Use $G_0 = 0.062$ exactly as stated in the problem.

**(g)** A prediction interval for one future observation must also include the variance of the new regression error:

$$se_{pred} = \sqrt{se(\widehat{U}_0)^2 + \hat{\sigma}^2}.$$

Both intervals have the same center, but the prediction interval is wider.

## Exercise 5.14: Normality of regressors and errors

Let $y_i = \beta_1 + \beta_2 x_i + e_i$, with independently sampled pairs $(y_i, x_i)$, $i = 1, \ldots, N$.

**(a)** Suppose $x_i$ is log-normal and $(e_i \mid x_i) \sim N(0, \sigma_e^2)$. If $W \sim N(\mu_W, \sigma_W^2)$, then $X = \exp(W)$ has $\mu_X = \exp(\mu_W + \sigma_W^2/2)$ and $\sigma_X^2 = [\exp(\sigma_W^2) - 1]\mu_X^2$.

- **(i)** Are the OLS estimators of $\beta_1$ and $\beta_2$ unbiased?
- **(ii)** Are they consistent?
- **(iii)** Are they normally distributed conditional on $\mathbf{x}$?
- **(iv)** Are their unconditional distributions normal?
- **(v)** Are $t$ tests exact in finite samples or only approximate?
- **(vi)** If $\mu_W = 0$, $\sigma_W^2 = 1$, and $x_i = \exp(w_i)$, find the asymptotic variance of $b_2$ in terms of $\sigma_e^2$ and $N$.

**(b)** Now suppose $x_i \sim N(0, 1)$ and the conditional error distribution is log-normal, with parameters $\mu_v$, $\sigma_v^2$ that do not depend on $x_i$. Its mean and variance are $\mu_e = \exp(\mu_v + \sigma_v^2/2)$ and $\sigma_e^2 = [\exp(\sigma_v^2) - 1]\mu_e^2$.

- **(i)** Rewrite the model as $y_i = \beta_1^* + \beta_2 x_i + e_i^*$, where $\beta_1^* = \beta_1 + \mu_e$ and $e_i^* = e_i - \mu_e$.
- **(ii)** Show that $E(e_i^* \mid x_i) = 0$ and $\text{var}(e_i^* \mid x_i) = \sigma_e^2$.
- **(iii)** Is $b_2$ unbiased for $\beta_2$?
- **(iv)** Is it consistent?
- **(v)** Is its distribution normal conditional on $\mathbf{x}$?
- **(vi)** Is its unconditional distribution normal?
- **(vii)** Are $t$ tests exact in finite samples or only approximate?
- **(viii)** Find the asymptotic variance of $b_2$ in terms of $\sigma_e^2$ and $N$.

## Hints for Exercise 5.14

A useful representation of the slope estimator is

$$b_2 = \beta_2 + \frac{\sum_i (x_i - \bar{x}) e_i}{\sum_i (x_i - \bar{x})^2}.$$

Use it to separate three questions: whether the estimator is centered on the true value, whether it converges to the true value, and whether its finite-sample distribution is normal.

**(a)(i)–(ii)** Conditional mean-zero errors are the key to unbiasedness. Random sampling, finite moments, positive regressor variance, and exogeneity are the ingredients for consistency. The regressor itself does not need to be normally distributed.

**(a)(iii)–(v)** Conditional on the observed $\mathbf{x}$, the OLS estimators are linear combinations of the errors. Ask what conditional normality of the errors therefore implies. For the unconditional distribution, remember that random regressor samples produce different conditional variances; averaging over them creates a mixture. Finally, exact finite-sample t inference depends on the distribution of the errors conditional on the regressors.

**(a)(vi)** In simple regression,

$$\text{Avar}(b_2) = \frac{\sigma_e^2}{N \,\text{var}(x_i)}.$$

Use the supplied log-normal formulas with $\mu_W = 0$ and $\sigma_W^2 = 1$ to find $\text{var}(x_i)$ before substituting.

**(b)(i)–(ii)** A log-normal error has a nonzero mean. Add and subtract $\mu_e$ so that the mean is absorbed into the intercept:

$$y_i = (\beta_1 + \mu_e) + \beta_2 x_i + (e_i - \mu_e).$$

Subtracting a constant changes the mean but not the variance. It also does not turn a log-normal variable into a normal variable.

**(b)(iii)–(vii)** Apply the same distinctions as in part (a). Mean-zero centered errors can support unbiasedness and consistency without being normal. However, a normal regressor cannot replace the conditional normal-error assumption needed for exact finite-sample coefficient normality and t tests. Consider what the central limit theorem provides as $N$ grows.

**(b)(viii)** Again use

$$\text{Avar}(b_2) = \frac{\sigma_e^2}{N \,\text{var}(x_i)},$$

now with the stated $\text{var}(x_i) = 1$.

### Exercise 5.15: Zero covariance and exogeneity

Let $y_i = \beta_1 + \beta_2 x_i + e_i$, with independently sampled pairs, $x_i \sim N(0,1)$, and $E(e_i \mid x_i) = c(x_i^2 - 1)$ for a constant $c$.

**(a)** Show that $E(e_i) = 0$.

**(b)** Using $\text{cov}(e_i, x_i) = E_x[(x_i - \mu_x)E(e_i \mid x_i)]$, show that $\text{cov}(e_i, x_i) = 0$.

**(c)** Is the OLS estimator of $\beta_2$ (i) unbiased and (ii) consistent? Explain.

## Hints for Exercise 5.15

For a standard normal variable, use $E(x_i) = 0$, $E(x_i^2) = 1$, and $E(x_i^3) = 0$.

**(a)** Apply the law of iterated expectations:

$$E(e_i) = E_x[E(e_i \mid x_i)].$$

Insert the stated conditional mean and use the second moment of a standard normal variable.

**(b)** Since $\mu_x = 0$, the supplied identity becomes

$$\operatorname{cov}(e_i, x_i) = E_x[x_i E(e_i \mid x_i)].$$

After substitution, expand the expectation into terms involving $E(x_i^3)$ and $E(x_i)$.

**(c)(i)** Do not confuse zero covariance with conditional mean zero. Use

$$E(b_2 \mid \mathbf{x}) - \beta_2 = \frac{\sum_i (x_i - \bar{x}) E(e_i \mid x_i)}{S_{xx}}$$

to examine conditional unbiasedness. For the unconditional expectation, also consider the symmetry of a normal regressor sample: replacing every $x_i$ by $-x_i$ changes the sign of the conditional-bias numerator but not $S_{xx}$.

**(c)(ii)** Write the estimation error as a ratio of sample averages. Under the usual moment conditions, its numerator converges to $\operatorname{cov}(x_i, e_i)$ and its denominator to $\operatorname{var}(x_i)$. Use your result from (b).

## Exercise 5.16: Prices of competing tuna brands

Weekly sales of brand A, $CANS$, are in thousands of cans. Define $RPRCE_B = 100PRICE_B/PRICE_A$ and $RPRCE_C = 100PRICE_C/PRICE_A$. The model is $\ln(CANS) = \beta_1 + \beta_2 RPRCE_B + \beta_3 RPRCE_C + e$.

**(a)** Assuming MR1-MR5, interpret $\beta_2$ and $\beta_3$ and discuss their expected signs.

With $N = 52$, the fitted equation is $\widehat{\ln(CANS)} = -2.724 + 0.0146 RPRCE_B + 0.02649 RPRCE_C$. Standard errors: $(0.582,\ 0.00548,\ 0.00544)$; $\hat{\sigma} = 0.5663$ and $\widehat{\text{cov}}(b_2, b_3) = -0.0000143$.

**(b)** At 10%, test whether a one-percentage-point increase in $RPRCE_B$ increases mean sales by 2.5%, against a two-sided alternative.

**(c)** Repeat (b) for $RPRCE_C$.

**(d)** At 10%, test $H_0 : \beta_2 = \beta_3$ against a two-sided alternative. Does this contradict (b)-(c)?

**(e)** Which competing brand seems the closer substitute for A? Why?

**(f)** Using the corrected predictor from Section 4.5.3, predict mean brand A sales at $(RPRCE_B, RPRCE_C)$ equal to (i) $(125, 100)$, (ii) $(111.11, 88.89)$, and (iii) $(100, 80)$.

**(g)** B costs 1 dollar and C costs 80 cents. Which price for A maximizes predicted revenue: 80 cents, 90 cents, or 1 dollar? Use (f).

## Hints for Exercise 5.16

There are $N - K = 52 - 3$ residual degrees of freedom. With only MR1–MR5 stated, treat the usual t procedures as approximations.

**(a)** This is a log-level model. A one-unit increase in a relative-price index changes sales by approximately $100\beta_j$ percent, holding the other relative price fixed. Think about substitution toward brand A when a competing brand becomes more expensive relative to A.

**(b)–(c)** Using the small-change approximation, a 2.5% effect corresponds to a coefficient of 0.025. For each slope, use

$$t = \frac{b_j - 0.025}{se(b_j)}$$

with a two-sided 10% test. For an exact 2.5% change on the exponentiated scale, the corresponding log change would be $\ln(1.025)$; state which convention you use.

**(d)** Rewrite the null as $H_0 : \beta_2 - \beta_3 = 0$. The standard error of the estimated difference is

$$se(b_2 - b_3) = \sqrt{se(b_2)^2 + se(b_3)^2 - 2\widehat{\text{cov}}(b_2, b_3)}.$$

Test the difference directly. Different conclusions in (b) and (c) do not by themselves prove that the two coefficients differ.

**(e)** Compare the two point estimates, but qualify the ranking using the formal equality test from (d).

**(f)** First calculate the fitted log value

$$\hat{\eta} = b_1 + b_2 RPRCE_B + b_3 RPRCE_C.$$

Under conditionally normal, homoskedastic log errors, use the corrected predictor

$$E(\widehat{CANS} \mid X) = \exp\left(\hat{\eta} + \frac{\hat{\sigma}^2}{2}\right).$$

Apply it separately to the three specified relative-price pairs. The predicted sales unit is thousands of cans.

**(g)** For each candidate price $P_A$, construct $RPRCE_B = 100P_B/P_A$ and $RPRCE_C = 100P_C/P_A$. Match the result to the corresponding sales prediction from (f), then calculate predicted revenue as price per can times predicted cans. This compares only the three stated prices, not every possible price or profit after costs.

### Exercise 5.17: Check the hand calculations in R

Use R to verify Exercise 5.1, parts **(c), (e), (f), (g), and (h)**. Use the seven observations on the Exercise 5.1 page.

Compare the software results with your hand calculations for:

- The three coefficient estimates.
- The estimated error variance.
- The correlation between the regressors.
- The standard error of $b_2$.
- SSE, SST, SSR, and $R^2$.

### Hints for Exercise 5.17

The student R worksheet already contains the seven observations as `exercise_51`. Estimate the multiple regression with

```
model_517 <- lm(y ~ x2 + x3, data = exercise_51)
summary(model_517)
```

Then use the following extraction commands and compare each result with your hand calculations from Exercise 5.1:

```
# Coefficient estimates
coef(model_517)

# Estimated error variance
summary(model_517)$sigma^2

# Correlation between the regressors
cor(exercise_51$x2, exercise_51$x3)

# Standard error of b2
coef(summary(model_517))["x2", "Std. Error"]

# Sums of squares and R-squared
SSE_517 <- sum(residuals(model_517)^2)
SST_517 <- sum((exercise_51$y - mean(exercise_51$y))^2)
SSR_517 <- SST_517 - SSE_517
c(SSE = SSE_517, SST = SST_517, SSR = SSR_517)
summary(model_517)$r.squared
```

If a software result differs from your hand calculation, first check the intercept, the residual degrees of freedom, and whether you used centered sums of squares.

### Exercise 5.18: Food, clothing and household size

Use `london5`. Budget shares satisfy $WFOOD = \beta_1 + \beta_2 \ln(TOTEXP) + e_F$ and $WCLOTH = \alpha_1 + \alpha_2 \ln(TOTEXP) + e_C$. $TOTEXP$ is total expenditure and $NK$ the number of children.

**(a)** A positive log-expenditure coefficient indicates a luxury; a negative coefficient indicates a necessity. What signs do you expect?

**(b)** Estimate both equations for one-child households. Discuss the estimates, significance, and any counterintuitive findings.

**(c)** At 1%, test $H_0 : \beta_2 \geq 0$ against $H_1 : \beta_2 < 0$. Why use this direction?

**(d)** At 1%, test $H_0 : \alpha_2 \geq 0$ against $H_1 : \alpha_2 < 0$. Why use this direction?

**(e)** Estimate both equations for two-child households. Compare 95% confidence intervals for the log-expenditure slopes across the one- and two-child groups. Do you expect equal slopes?

**(f)** Using all observations, estimate $WFOOD = \gamma_1 + \gamma_2 \ln(TOTEXP) + \gamma_3 NK + \gamma_4 NK \ln(TOTEXP) + e_F$ and $WCLOTH = \delta_1 + \delta_2 \ln(TOTEXP) + \delta_3 NK + \delta_4 NK \ln(TOTEXP) + e_C$. Test your conjectures from (e) at 5% (95% confidence).

**(g)** Compare the marginal log-expenditure effects for food at $NK = 1, 2$ from the separate regressions and the interaction model.

### Hints for Exercise 5.18

The student R worksheet downloads `london5` in `START UP`. Use `names(london5)` and `head(london5)` to inspect the variables before estimating the models.

**(a)** Think about how each budget **share** changes when total expenditure rises. A necessity normally takes a smaller share of the budget as households become richer, whereas a luxury may take a larger share. Translate these ideas into expected signs for the coefficients on $\ln(TOTEXP)$.

**(b)** First select the one-child households, then estimate the two equations:

```
one_child <- london5 |>
  filter(nk == 1)

food_one <- lm(wfood ~ log(totexp), data = one_child)
clothing_one <- lm(wcloth ~ log(totexp), data = one_child)

summary(food_one)
summary(clothing_one)
```

Use the signs, sizes, standard errors, and p-values when discussing the results. Since the dependent variables are shares, distinguish percent from percentage points.

**(c)** This is a lower-tail test. Obtain the slope's t-statistic and residual degrees of freedom from the fitted food model. The one-sided p-value is

$$P(T_{N-K} \leq t) = F_{t_{N-K}}(t).$$

In R, `pt(t_value, df)` gives this lower-tail probability. Reject at 1% when the one-sided p-value is below 0.01. Check the sign of the estimate before drawing the conclusion.

**(d)** Repeat the procedure from (c) for the clothing model. Do not automatically divide a two-sided p-value by two without first checking whether the estimated coefficient points in the direction specified by $H_1$.

**(e)** Create a two-child subsample and estimate the same two regressions. Use `confint(model, "log(totexp)", level = 0.95)` for each slope. Comparing separate confidence intervals is descriptive; overlapping intervals do not by themselves constitute a formal test that the two slopes are equal.

**(f)** With all observations, `*` includes both main effects and the interaction:

```
food_interaction <- lm(wfood ~ log(totexp) * nk, data = london5)
clothing_interaction <- lm(wcloth ~ log(totexp) * nk, data = london5)
```

In each model, test whether the coefficient on `log(totexp):nk` is zero. A zero interaction coefficient means that the log-expenditure slope does not change with the number of children.

**(g)** In the food interaction model, the marginal effect of $\ln(TOTEXP)$ for a household with $NK = n$ children is

$$\frac{\partial E(WFOOD \mid \mathbf{X})}{\partial \ln(TOTEXP)} = \gamma_2 + n\gamma_4.$$

Evaluate this expression at $n = 1$ and $n = 2$, and compare the resulting slopes with those from the two separate food regressions.

### Exercise 5.19: Food expenditure elasticity

Use all observations in `london5` and the model $WFOOD = \beta_1 + \beta_2 \ln(TOTEXP) + e_F$. The elasticity of food expenditure with respect to total expenditure is $\varepsilon = 1 + \beta_2/[\beta_1 + \beta_2 \ln(TOTEXP)]$.

**(a)** Show that $\beta_2 < 0$ if and only if $\varepsilon < 1$ when the mean food share is positive. Relate this to a necessity.

**(b)** Estimate the equation and comment on the results.

**(c)** Find point estimates and 95% confidence intervals for the mean food budget share at $TOTEXP$ equal to (i) 50 (5th percentile), (ii) 90 (median), and (iii) 170 (95th percentile).

**(d)** Find point estimates and 95% confidence intervals for $\varepsilon$ at the same three expenditure levels.

**(e)** How do the mean share, elasticity and reliability of estimation change with total expenditure?

### Hints for Exercise 5.19

Download the textbook data in `START UP` if they are not already loaded:

```
load(url("https://www.principlesofeconometrics.com/poe5/data/rdata/london5.rdata"))
```

**(a)** Let $w = \beta_1 + \beta_2 \ln(TOTEXP)$ denote the positive mean food share. Then

$$\varepsilon - 1 = \frac{\beta_2}{w}.$$

When $w > 0$, the sign of the right-hand side is the sign of $\beta_2$. Use this to connect a negative slope with an elasticity below one.

**(b)** Estimate the model using all households:

```
food_519 <- lm(wfood ~ log(totexp), data = london5)
summary(food_519)
```

Discuss the slope's sign, magnitude, uncertainty, and $R^2$. Remember that `wfood` is a budget share.

**(c)** Create a small data frame with the three expenditure values and request confidence intervals for the conditional mean:

```
budgets_519 <- tibble(totexp = c(50, 90, 170))
predict(food_519, newdata = budgets_519, interval = "confidence")
```

These are confidence intervals for mean food shares, not prediction intervals for individual households.

**(d)** The elasticity is a nonlinear function of both coefficients. Use the full covariance matrix through `car::deltaMethod()`. For example:

```
car::deltaMethod(
  food_519,
  "1 + b2 / (b1 + b2 * log(50))",
  parameterNames = c("b1", "b2")
)
```

Repeat for 90 and 170. The resulting intervals are delta-method approximations.

**(e)** Compare the fitted mean shares, elasticity estimates, and interval widths across the three expenditure levels. A smaller fitted share can make the elasticity more sensitive to coefficient uncertainty.

### Exercise 5.20: An alternative slope estimator

For $y_i = \beta_1 + \beta_2 x_i + e_i$, assume MR1-MR6 and even $N$. Sort observations by increasing $x$. Let $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ be means for the lower and upper halves. Consider $\widehat{\beta}_{2,mean} = (\bar{y}_2 - \bar{y}_1)/(\bar{x}_2 - \bar{x}_1)$.

**(a)** Show that this estimator is unbiased.

**(b)** Derive its conditional variance given $\mathbf{x}$.

**(c)** Write an expression for its unconditional variance.

**(d)** Under what conditions is it consistent?

**(e)** Generate $x_i \sim U(0, 10)$ for $N = 100, 500, 1000$ and, if possible, 5000. Set $\sigma^2 = 1000$. For each $N$, compute: (i) the conditional variances of OLS $b_2$ and $\widehat{\beta}_{2,mean}$; (ii) estimates of $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$. Here $s_x^2 = N^{-1}\sum_i (x_i - \bar{x})^2$ uses divisor $N$.

**(f)** Compare the magnitudes in (e) and their changes with $N$. Does the alternative estimator appear consistent?

**(g)** Explain the large-sample limits 0.12 for the inverse-variance quantity and 0.16 for the squared-gap quantity in (e)(ii). Hint: $\text{var}[U(a, b)] = (b - a)^2/12$.

**(h)** Would the estimator be consistent if observations were assigned randomly to the two halves instead of sorted by $x$? Explain.

### Hints for Exercise 5.20

Let $D = \bar{x}_2 - \bar{x}_1$. Because each half contains $N/2$ observations, keep track of the variance of each group mean error.

**(a)** Substitute $y_i = \beta_1 + \beta_2 x_i + e_i$ into both group means. The intercept cancels when the means are subtracted. Conditional on the sorted regressor values, use $E(\bar{e}_2 - \bar{e}_1 \mid \mathbf{x}) = 0$.

**(b)** The two group mean errors are based on disjoint observations. Use

$$\operatorname{var}(\bar{e}_j \mid \mathbf{x}) = \frac{\sigma^2}{N/2}$$

for each half, add their variances, and divide by $D^2$.

**(c)** Apply the law of total variance. Since the conditional mean of the estimator is constant, only the expectation of the conditional variance remains.

**(d)** A useful sufficient condition is that $ND^2 \to \infty$ in probability. Ask whether the sorted lower- and upper-half means remain separated as the sample grows.

**(e)** The following structure computes the requested quantities without simulating $y$, because the conditional variances depend only on $x$:

```
compare_520 <- function(N) {
  x <- sort(runif(N, 0, 10))
  lower <- seq_len(N / 2)
  D <- mean(x[-lower]) - mean(x[lower])
  sx2 <- mean((x - mean(x))^2)

  c(
    ols_variance = 1000 / (N * sx2),
    split_variance = 4000 / (N * D^2),
    inverse_variance = 1 / sx2,
    inverse_gap = 4 / D^2
  )
}
```

Run it for the requested sample sizes. To estimate expectations, repeat each experiment many times with `replicate()` and average the returned quantities.

**(f)** Compare which estimator uses more of the available variation in $x$. Then examine whether both estimated variances decline as $N$ increases.

**(g)** For $U(0, 10)$, the population variance is $(10 - 0)^2/12$. After sorting, the lower half approaches a uniform distribution on $(0, 5)$ and the upper half one on $(5, 10)$. Use their limiting means to find the limiting gap $D$ and evaluate $1/\operatorname{var}(x)$ and $4/D^2$.

**(h)** Without sorting, both halves are drawn from the same population, so both group means approach the same value. Consider what this does to $D$, the denominator of the estimator, and the condition $ND^2 \to \infty$.

## Exercise 5.21: Rainfall and technological change

Using `toody5`, estimate $Y_t = \beta_1 + \beta_2 TREND_t + \beta_3 RAIN_t + \beta_4 RAIN_t^2 + \beta_5 RAIN_t TREND_t + e_t$. Yield $Y$ is tonnes per hectare in Toodyay, Western Australia. $TREND = 0, 0.1, \ldots, 4.7$ for 1950-1997. Growing-season rainfall is in decimeters (the book uses approximately 4 inches per decimeter).

**(a)** Report coefficient estimates, standard errors, $t$ statistics, and p-values in a table.

**(b)** Which estimates differ from zero at (i) 5% and (ii) 10%?

**(c)** Do the coefficients have the expected signs? Explain, considering that technological progress may produce drought-resistant varieties.

**(d)** Find point estimates and 95% confidence intervals for the marginal effect of rainfall in (i) 1959, with $RAIN = 2.98$, and (ii) 1995, with $RAIN = 4.797$. Discuss.

**(e)** Find point estimates and 95% confidence intervals for rainfall that maximizes expected yield in (i) 1959 and (ii) 1995. Discuss.

### Hints for Exercise 5.21

The student R worksheet downloads `toody5` in `START UP`. Use `head(toody5)` and `names(toody5)` to inspect the data. The values of `trend` are 0.9 in 1959 and 4.5 in 1995.

**(a)** Estimate the model with a squared rainfall term and an interaction:

```
model_521 <- lm(
  y ~ trend + rain + I(rain^2) + trend:rain,
  data = toody5
)

coef(summary(model_521))
```

`I(rain^2)` tells R to use the numerical square of rainfall. The coefficient table contains the estimates, standard errors, t-statistics, and p-values.

**(b)** The reported p-values are for two-sided tests of $H_0 : \beta_k = 0$. Compare each p-value first with 0.05 and then with 0.10. A coefficient can fail to be significant at 5% but still be significant at 10%.

**(c)** Consider the model as a curved rainfall response that may change over time. A positive coefficient on $RAIN$ and a negative coefficient on $RAIN^2$ allow diminishing returns and eventually a decline in yield when rainfall is very high. The rainfall–trend interaction shows whether the marginal rainfall effect changes as technology develops. Interpret combinations of coefficients, not each term in isolation.

**(d)** Differentiate the conditional mean with respect to rainfall:

$$\frac{\partial E(Y_t \mid \mathbf{X}_t)}{\partial RAIN_t} = \beta_3 + 2\beta_4 RAIN_t + \beta_5 TREND_t.$$

Insert $(RAIN, TREND) = (2.98, 0.9)$ for 1959 and $(4.797, 4.5)$ for 1995. The expression is a linear combination of coefficients. You can obtain its estimate and standard error with `car::deltaMethod()`:

```
car::deltaMethod(
  model_521,
  "b3 + 2*b4*2.98 + b5*0.9",
  parameterNames = c("b1", "b2", "b3", "b4", "b5")
)
```

Change the rainfall and trend values for 1995. Form each interval as

$$\widehat{m} \pm t_{0.975,\, N-K}\, se(\widehat{m}).$$

**(e)** Expected yield is maximized where the marginal rainfall effect from part (d) equals zero. At trend value $T$, solve for rainfall:

$$RAIN^*(T) = -\frac{\beta_3 + \beta_5 T}{2\beta_4}.$$

Check the sign of $\beta_4$ to determine whether the turning point is a maximum. Estimate the nonlinear expression and its delta-method standard error with, for example,

```r
car::deltaMethod(
  model_521,
  "-(b3 + b5*0.9)/(2*b4)",
  parameterNames = c("b1", "b2", "b3", "b4", "b5")
)
```

Replace 0.9 with 4.5 for 1995, then construct the approximate 95% intervals. Because the turning point is a nonlinear ratio of coefficients, these intervals use a large-sample delta-method approximation.

## Exercise 5.22: Climate and technological change

Use `toody5` and the model $Y_t = \beta_1 + \beta_2 TREND_t + \beta_3 RAIN_t + \beta_4 RAIN_t^2 + \beta_5 RAIN_t TREND_t + e_t$. Yield is tonnes per hectare; rainfall is in decimeters; $TREND = 0, 0.1, \ldots, 4.7$ for 1950-1997.

**(a)** Report estimates, standard errors, $t$ statistics, and p-values.

**(b)** For 1974 ($TREND = 2.4$, $RAIN = 4.576$), test at 5% that extra rainfall does not increase expected yield against a positive effect.

**(c)** Hold rainfall at its median, 3.8355. Find a point estimate and 95% confidence interval for the expected yield improvement attributable to technological change from 1960 to 1995.

**(d)** Estimate $RAIN = \alpha_1 + \alpha_2 TREND + e$. At 5%, test no decline in mean rainfall against declining mean rainfall.

**(e)** Predict mean rainfall in 1960 and 1995 using (d).

**(f)** Hold $TREND_{1995} = TREND_{1960}$, so there is no technological change. Using (e), estimate the decline in mean yield attributable to changing rainfall from 1960 to 1995.

**(g)** Hold $E(RAIN_{1995}) = E(RAIN_{1960})$, so there is no rainfall change. Estimate the increase in mean yield attributable to technology.

**(h)** Compare the estimates from (c), (f), and (g).

### Hints for Exercise 5.22

Continue with `toody5`, downloaded from the textbook website for Exercise 5.21. The trend values are 1 in 1960, 2.4 in 1974, and 4.5 in 1995.

**(a)** Estimate the same interaction model as in Exercise 5.21:

```
model_522 <- lm(
  y ~ trend + rain + I(rain^2) + trend:rain,
  data = toody5
)
coef(summary(model_522))
```

**(b)** At rainfall $R$ and trend $T$, the marginal rainfall effect is

$$m(R, T) = \beta_3 + 2R\beta_4 + T\beta_5.$$

Evaluate it at $(R, T) = (4.576, 2.4)$. Obtain its standard error with `car::deltaMethod()`, then calculate $t = \widehat{m}/se(\widehat{m})$. This is an upper-tail test because the alternative says that rainfall increases yield.

**(c)** Holding rainfall at $R = 3.8355$, the technological change from 1960 to 1995 is

$$(4.5 - 1)(\beta_2 + R\beta_5).$$

This is a linear combination of coefficients. Use `deltaMethod()` or the full coefficient covariance matrix, then form a 95% confidence interval.

**(d)** Estimate the rainfall trend with

```
rain_model_522 <- lm(rain ~ trend, data = toody5)
summary(rain_model_522)
```

Test $H_0 : \alpha_2 \geq 0$ against $H_1 : \alpha_2 < 0$. Use the lower-tail p-value `pt(t_value, df)` rather than the two-sided p-value printed by `summary()`.

**(e)** Predict the two conditional mean rainfall values:

```
rain_means_522 <- predict(
  rain_model_522,
  newdata = tibble(trend = c(1, 4.5))
)
```

**(f)** Construct two cases with trend fixed at 1 and rainfall equal to the two predicted means from (e). Predict yield for both rows and subtract the 1960 prediction from the 1995-rainfall prediction.

**(g)** Now hold rainfall fixed at its predicted 1960 mean while changing trend from 1 to 4.5. Again predict both rows and take the difference.

**(h)** Compare the three counterfactual changes carefully: part (c) holds rainfall at the median, part (f) changes only rainfall at 1960 technology, and part (g) changes only technology at 1960 mean rainfall. Because the model has an interaction, separate changes need not add exactly to the change when both inputs move.

### Exercise 5.23: Quantity discounts and quality premia

The `cocaine` data contain 56 sales in northeastern California, 1984-1991. $PRICE$ is dollars per gram, $QUANT$ grams sold, $QUAL$ percentage purity, and $TREND = 1, \ldots, 8$ for 1984-1991. Consider $PRICE = \beta_1 + \beta_2 QUANT + \beta_3 QUAL + \beta_4 TREND + e$.

**(a)** What signs do you expect for $\beta_2$, $\beta_3$, and $\beta_4$?

**(b)** Estimate the model, report the results and interpret the coefficients. Do the signs match your expectations?

**(c)** What fraction of price variation is jointly explained by quantity, quality and time?

**(d)** Sellers may accept a lower unit price for larger sales because fewer transactions reduce their exposure to detection. State suitable null and alternative hypotheses and test for a quantity discount.

**(e)** Test that quality has no effect on expected price against a positive quality premium.

**(f)** What is the estimated average annual change in price? Suggest an explanation for its direction.

### Hints for Exercise 5.23

Download the textbook data directly in `START UP` if needed:

```
load(url("https://www.principlesofeconometrics.com/poe5/data/rdata/cocaine.rdata"))
head(cocaine)
```

The variables are lowercase in R. Price is dollars per gram, quantity is grams per sale, and quality is percentage purity.

**(a)** A quantity discount suggests a negative coefficient on `quant`. Greater purity should command a positive quality premium. The trend sign is ambiguous because supply, demand, enforcement, and distribution conditions may all change over time.

**(b)** Estimate

```
model_523 <- lm(price ~ quant + qual + trend, data = cocaine)
coef(summary(model_523))
```

Interpret every slope while holding the other included variables constant and retain the units of dollars per gram.

**(c)** Extract `summary(model_523)$r.squared`. Interpret it as the fraction of sample price variation around the mean jointly explained by quantity, quality, and trend.

**(d)** Translate "quantity discount" into $H_0 : \beta_2 \geq 0$ against $H_1 : \beta_2 < 0$. The reported t-statistic tests the boundary value zero. Obtain the lower-tail p-value with `pt(t_value, df.residual(model_523))`.

**(e)** For a positive quality premium, use $H_0 : \beta_3 \leq 0$ against $H_1 : \beta_3 > 0$. Obtain the upper-tail p-value with `pt(t_value, df.residual(model_523), lower.tail = FALSE)`.

**(f)** The coefficient on `trend` is the estimated average annual change in price per gram, holding quantity and quality fixed. Its sign describes a time pattern but does not identify which economic mechanism caused it.

### Exercise 5.24: House size and age categories

The `collegetown` data cover 500 Baton Rouge houses sold in 2009-2013. $PRICE$ is thousands of dollars, $SQFT$ hundreds of square feet, and $AGE$ a category from 1 (newest) to 11 (oldest). Use the age definitions in the publisher's `collegetown.def` file. Estimate $PRICE = \beta_1 + \beta_2 SQFT + \beta_3(SQFT \times AGE) + e$.

**(a)** Report results. Do the coefficients differ significantly from zero?

**(b)** Derive the marginal effects with respect to $SQFT$ and $AGE$. Interpret the coefficients. What assumptions follow from treating the ordered age categories this way?

**(c)** Find point estimates and 95% confidence intervals for the size effect at ages (i) 5, (ii) 20, and (iii) 40 years, using the correct age codes. How does the effect change with age?

**(d)** For houses of (i) 1500, (ii) 3000, and (iii) 4500 square feet, test at 5% that moving one age category higher reduces expected price by 6000 dollars, against a two-sided alternative. Discuss.

**(e)** Find a 95% prediction interval for a 60-year-old, 2500-square-foot house. Identify the four observations in its age category with floor space from 2450 to 2550 square feet. What were their prices? How many lie inside the interval? Assess the model's forecasting usefulness.

### Hints for Exercise 5.24

Download the data directly in `START UP`:

```
load(url("https://www.principlesofeconometrics.com/poe5/data/rdata/collegetown.rdata"))
head(collegetown)
```

`price` is measured in thousands of dollars and `sqft` in hundreds of square feet. `age` is a category code, not the house's exact age in years.

**(a)** Estimate exactly the model stated in the exercise:

```
model_524 <- lm(price ~ sqft + sqft:age, data = collegetown)
coef(summary(model_524))
```

Do not use `sqft * age`, because that would also add an age main effect and estimate a different model.

**(b)** The size effect at age code $A$ and the effect of moving one category higher at size $S$ are

$$\frac{\partial E(PRICE \mid \mathbf{X})}{\partial SQFT} = \beta_2 + A\beta_3, \qquad \Delta_{AGE} E(PRICE \mid \mathbf{X}) = S\beta_3.$$

Treat the second as a discrete category change. Using numeric category codes assumes the same price change for every one-step category increase at a fixed size, even though the underlying age bands have unequal widths.

**(c)** The relevant age codes are 3 for 5 years, 6 for 20 years, and 8 for 40 years. At code $A$, estimate $b_2 + Ab_3$ and use

$$\widehat{\text{var}}(b_2 + Ab_3) = \widehat{\text{var}}(b_2) + A^2\widehat{\text{var}}(b_3) + 2A\widehat{\text{cov}}(b_2, b_3).$$

Extract the required variances and covariance with `vcov(model_524)` and form the three 95% intervals.

**(d)** Convert 1500, 3000, and 4500 square feet to `sqft` values 15, 30, and 45. At size $S$, the estimated category effect is $Sb_3$ and its standard error is $|S|se(b_3)$. Price is in thousands of dollars, so a \$6000 decline corresponds to the null value $-6$. Use

$$t = \frac{Sb_3 - (-6)}{|S|se(b_3)}$$

for each two-sided 5% test.

**(e)** A 60-year-old house is in category 10, and 2500 square feet means `sqft = 25`. Request an interval for an individual house:

```
prediction_524 <- predict(
  model_524,
  newdata = tibble(sqft = 25, age = 10),
  interval = "prediction"
)
```

Find the comparison houses with

```r
nearby_524 <- collegetown |>
  filter(age == 10, sqft >= 24.5, sqft <= 25.5) |>
  select(price, sqft, age)
```

Compare each observed price with the lower and upper prediction limits. When assessing usefulness, consider interval width as well as the number of covered observations.

### Exercise 5.25: House prices in a nonlinear log model

Use all 500 observations in `collegetown`. $PRICE$ is thousands of dollars and $SQFT$ hundreds of square feet. Estimate $\ln(PRICE) = \beta_1 + \beta_2 SQFT + \beta_3\sqrt{SQFT} + e$. Assume MR1-MR6, including $e \mid SQFT \sim N(0, \sigma^2)$.

**(a)** Report results and assess coefficient significance.

**(b)** Derive $\partial E[\ln(PRICE) \mid SQFT]/\partial SQFT$. Discuss its form and the expected signs of $\beta_2$ and $\beta_3$.

**(c)** Find and interpret point estimates and 95% confidence intervals for that derivative at 1500, 3000, and 4500 square feet.

**(d)** Show that $\partial E(PRICE \mid SQFT)/\partial SQFT = S \times C$, where $S = [\beta_2 + \beta_3/(2\sqrt{SQFT})]\exp(\beta_1 + \beta_2 SQFT + \beta_3\sqrt{SQFT})$ and $C = \exp(\sigma^2/2)$. A hat denotes replacement by estimated parameters.

**(e)** Estimate $S \times C$ for each size in (c).

**(f)** Find standard errors for $\hat{S}$ at each size, initially avoiding the additional uncertainty in $\hat{C}$.

**(g)** Using estimates from (e) and standard errors from (f), test at 5% whether the price derivative equals 9 against a two-sided alternative, at each size. State your conclusions.

**(h), optional.** Given $\text{var}(\hat{\sigma}^2) = 2\sigma^4/(N-K)$, estimate $\text{var}(\hat{C})$. The book states that $\hat{S}$ and $\hat{C}$ are independent. Use $\widehat{\text{var}}(\hat{S}\hat{C}) = \hat{S}^2\widehat{\text{var}}(\hat{C}) + \hat{C}^2\widehat{\text{var}}(\hat{S}) + \widehat{\text{var}}(\hat{C})\widehat{\text{var}}(\hat{S})$ to find standard errors for the product. Compare with (f). Would the tests in (g) change?

### Hints for Exercise 5.25

Start by estimating the log-price model:

```
model_525 <- lm(log(price) ~ sqft + sqrt(sqft), data = collegetown)
summary(model_525)
```

**(a)** Use the coefficient table to report estimates, standard errors, $t$ statistics and p-values. Remember that the regressors `sqft` and `sqrt(sqft)` belong to the same nonlinear relationship; interpreting either coefficient in isolation can therefore be misleading.

**(b)** Differentiate the conditional log-price function with respect to $SQFT$:

$$m(x) = \frac{\partial E[\ln(PRICE) \mid x]}{\partial x} = \beta_2 + \frac{\beta_3}{2\sqrt{x}}.$$

Study how this derivative changes with $x$. The signs of $\beta_2$ and $\beta_3$ determine whether the marginal effect rises or falls with house size.

**(c)** Because `sqft` is measured in hundreds of square feet, use $x = 15, 30, 45$. At a chosen $x$, the estimated derivative is $b_2 + b_3/(2\sqrt{x})$. Its standard error must include the covariance between $b_2$ and $b_3$. One convenient approach is:

```
library(car)
deltaMethod(
  model_525,
  "b2 + b3 / (2 * sqrt(x))",
  parameterNames = c("b1", "b2", "b3"),
  constants = c(x = 15)
)
```

Repeat for the other two values and form each interval with the appropriate $t$ critical value.

**(d)** Under the normality assumption, $E(PRICE \mid x) = \exp(\beta_1 + \beta_2 x + \beta_3\sqrt{x} + \sigma^2/2)$. Differentiate this expression. The chain rule separates the result into the stated $S$ and the retransformation factor $C = \exp(\sigma^2/2)$.

**(e)** Obtain $\hat{\sigma}$ from `summary(model_525)$sigma`, calculate $\hat{C} = \exp(\hat{\sigma}^2/2)$, and multiply it by $\hat{S}$ at each size. The result is in thousands of dollars per 100 square feet.

**(f)** Treat $C$ as fixed for now. Write $S$ as a function of $b_1, b_2, b_3$ at each chosen $x$ and use `deltaMethod()` to obtain $se(\hat{S})$. Multiplying by $\hat{C}$ gives the corresponding approximate standard error for $\hat{S}\hat{C}$ when uncertainty in $\hat{C}$ is ignored.

**(g)** For each house size, use the two-sided statistic

$$t = \frac{\hat{S}\hat{C} - 9}{\hat{C}\, se(\hat{S})}.$$

Compare $|t|$ with the two-sided 5% critical value, or calculate its p-value with `2 * pt(-abs(t), df = df.residual(model_525))`.

**(h)** Apply the delta method to $C = \exp(\sigma^2/2)$:

$$\widehat{\text{var}}(\hat{C}) = \hat{C}^2 \frac{\hat{\sigma}^4}{2(N-K)}.$$

Insert this and $\widehat{\text{var}}(\hat{S})$ into the product- variance formula supplied in the question. Recalculate the test statistics and check whether any 5% decision changes.

### Exercise 5.26: Growth, inflation and presidential votes

Use all observations in `fair5`. $VOTE$ is the Democratic percentage of the two-party vote. $GROWTH$ and $INFLAT$ are economic growth and inflation multiplied by +1 under Democratic incumbency and -1 under Republican incumbency (see the definitions accompanying `fair5`). Estimate $VOTE = \beta_1 + \beta_2 GROWTH + \beta_3 INFLAT + e$.

**(a)** Report results. Are $\beta_2$ and $\beta_3$ different from zero at 10%? Did you use one- or two-tailed tests? Explain.

**(b)** With inflation 3% and Democratic incumbency, predict both parties' vote shares for economic growth of (i) -2%, (ii) 0%, and (iii) 3%.

**(c)** Repeat (b) under Republican incumbency.

**(d)** Do the predictions suggest a baseline popular-vote advantage for Democrats or Republicans?

**(e)** Compare two scenarios, both with economic growth and inflation of 2%: (1) Democratic incumbency and (2) Republican incumbency. At 5%, test whether the expected Democratic share in scenario 1 equals the expected Republican share in scenario 2.

### Hints for Exercise 5.26

Estimate the model with:

```
model_526 <- lm(vote ~ growth + inflat, data = fair5)
summary(model_526)
```

**(a)** "Different from zero" calls for two-sided tests unless a directional alternative was specified before seeing the estimates. At the 10% level, compare each p-value with 0.10.

**(b)** Under Democratic incumbency, the signed variables have the same signs as the stated economic outcomes. Construct the three rows and use `predict()`:

```
dem_526 <- tibble(growth = c(-2, 0, 3), inflat = 3)
dem_526$dem_vote <- predict(model_526, newdata = dem_526)
dem_526$rep_vote <- 100 - dem_526$dem_vote
```

**(c)** Under Republican incumbency, reverse the signs of both economic variables. The model still predicts the Democratic vote share, so subtract the prediction from 100 to obtain the Republican share.

**(d)** The intercept is the expected Democratic share when both signed economic variables equal zero. Compare $b_1$ with 50 to identify the fitted baseline advantage. Also inspect whether the predictions in (b) and (c) show the same pattern.

**(e)** Write the Democratic share under Democratic incumbency as $b_1 + 2b_2 + 2b_3$. Under Republican incumbency, the predicted Democratic share is $b_1 - 2b_2 - 2b_3$, so the Republican share is its complement. Equating the two incumbent-party shares simplifies to a restriction involving only the intercept. Test that restriction with

$$t = \frac{b_1 - 50}{se(b_1)}$$

using a two-sided 5% test.

### Exercise 5.27: Painting age, area and auction prices

Use `ashcan_small`, keeping only sold paintings ($SOLD = 1$). $RHAMMER$ is sale price in thousands of dollars, $YEARS_OLD$ time from completion to auction, and $INCHSQ$ area in square inches. Create $INCHSQ10 = INCHSQ/10$.

**(a)** Estimate and report $RHAMMER = \beta_1 + \beta_2 YEARS_OLD + \beta_3 INCHSQ10 + e$.

**(b)** Find a point estimate and 95% confidence interval for expected yearly appreciation in price.

**(c)** Find a point estimate and 95% confidence interval for the extra value associated with 10 additional square inches.

**(d)** Add $INCHSQ10^2$ and re-estimate. Report results and explain why a squared term could be useful.

**(e)** How much does this change the interval in (b)?

**(f)** Using the quadratic model, find 95% confidence intervals for the marginal value of 10 additional square inches at areas (i) 50, (ii) 250, and (iii) 900 square inches. Discuss the pattern.

**(g)** Find a 95% confidence interval for the area that maximizes price.

**(h)** Find a 95% confidence interval for expected price of a 75-year-old, 100-square-inch painting.

**(i)** How long must a 100-square-inch painting be kept before its expected price becomes positive?

### Hints for Exercise 5.27

Prepare the sample and rescaled area variable first:

```
art_527 <- ashcan_small |>
  filter(sold == 1) |>
  mutate(inchsq10 = inchsq / 10)
```

**(a)** Estimate `lm(rhammer ~ years_old + inchsq10, data = art_527)`. The coefficients are measured in thousands of dollars; the area coefficient is the fitted change associated with 10 additional square inches.

**(b)–(c)** The age and area coefficients are already the requested marginal effects in the linear model. Use `confint()` to obtain their 95% intervals.

**(d)** Estimate

```
model_527q <- lm(
  rhammer ~ years_old + inchsq10 + I(inchsq10^2),
  data = art_527
)
```

The squared term permits the marginal value of area to depend on the current area rather than remain constant.

**(e)** Compare the `years_old` interval from the two models. Do not compare only its point estimate; also note whether its precision changes.

**(f)** At $x = INCHSQ10$, the marginal value of 10 additional square inches is

$$\beta_3 + 2\beta_4 x.$$

Use $x = 5, 25, 90$. Apply `deltaMethod()` to `"b3 + 2*b4*x"`, supplying the chosen value as a constant, and form the three 95% intervals.

**(g)** The fitted turning point in the rescaled area is $-b_3/(2b_4)$. Multiply by 10 to return to square inches. Use `deltaMethod()` on `"-10*b3/(2*b4)"` to obtain its standard error and confidence interval. Check that the quadratic coefficient makes this point a maximum rather than a minimum.

**(h)** Here `years_old = 75` and `inchsq10 = 10`. Use

```
predict(
  model_527q,
  newdata = tibble(years_old = 75, inchsq10 = 10),
  interval = "confidence"
)
```

A confidence interval is appropriate because the question asks about an expected price, not the price of one individual painting.

**(i)** Set the fitted quadratic-model price equal to zero at `inchsq10 = 10` and solve for age:

$$YEARS_OLD = -\frac{b_1 + 10b_3 + 10^2 b_4}{b_2}.$$

If the question is interpreted as additional time beyond a stated current age, subtract that current age from the threshold.

### Exercise 5.28: Painting prices in a log model

Use `ashcan_small`, keeping $SOLD = 1$. $RHAMMER$ is price in thousands of dollars, $YEARS_OLD$ the painting's age, and $INCHSQ$ its area. Create $INCHSQ10 = INCHSQ/10$.

**(a)** Estimate and report $\ln(RHAMMER) = \beta_1 + \beta_2 YEARS_OLD + \beta_3 INCHSQ10 + e$.

**(b)** Find yearly percentage appreciation and a 95% confidence interval.

**(c)** At 5%, test that 10 additional square inches increase value by 2% or less against an increase greater than 2%.

**(d)** Add $INCHSQ10^2$, re-estimate, report results and explain why the additional term may be appropriate.

**(e)** How much does this change the interval in (b)?

**(f)** Repeat (c) for areas (i) 50, (ii) 250, and (iii) 900 square inches. What do you observe?

**(g)** Find a 95% confidence interval for the area that maximizes price.

**(h)** Find a 95% interval for the price quantity requested in the book for a 75-year-old, 100-square-inch painting. Use $\exp\{E[\ln(RHAMMER) \mid \widehat{YEARS_OLD} = 75, INCHSQ10 = 10]\}$ and its standard error. Explain what price quantity this estimator measures.

### Hints for Exercise 5.28

Use the sold paintings and create `inchsq10` exactly as in Exercise 5.27.

**(a)** Estimate `lm(log(rhammer) ~ years_old + inchsq10, data = art_528)` and report the usual coefficient table.

**(b)** A convenient approximate interpretation is $100b_2$ percent per year, with interval endpoints obtained by multiplying the coefficient interval by 100. For an exact percentage transformation, use $100[\exp(b_2) - 1]$ and apply the same transformation to the endpoints.

**(c)** Ten additional square inches correspond to a one-unit increase in `inchsq10`. For the approximate percentage interpretation, test $H_0 : \beta_3 \leq 0.02$ against $H_1 : \beta_3 > 0.02$ with

$$t = \frac{b_3 - 0.02}{se(b_3)}.$$

For an exact 2% threshold, replace 0.02 by $\ln(1.02)$. Use an upper-tail p-value.

**(d)** Add `I(inchsq10^2)` to the model. The quadratic allows percentage price gains from extra area to differ between small and large paintings.

**(e)** Compare the 95% interval for `years_old` across the linear and quadratic area specifications.

**(f)** In the quadratic model, the marginal log-price effect at area $x$ is $b_3 + 2b_4x$. Use $x = 5, 25, 90$ and `deltaMethod()` to find its estimate and standard error. For each area, test the expression against 0.02 using an upper-tail test.

**(g)** Convert the fitted turning point back to square inches:

$$A^* = -10\frac{b_3}{2b_4}.$$

Use the delta method for its interval, and confirm from the sign of $b_4$ whether it is a maximum.

**(h)** First obtain the fitted log price and its standard error:

```
pred_528 <- predict(
  model_528q,
  newdata = tibble(years_old = 75, inchsq10 = 10),
  se.fit = TRUE
)
```

If $g = \exp(E[\ln(RH\widehat{AMMER}) \mid X])$, then the delta-method standard error is approximately $g\, se(E[\ln(RH\widehat{AMMER}) \mid X])$. This quantity is the conditional geometric mean and, under normal log errors, the conditional median. It is not the conditional arithmetic mean unless a retransformation factor is included.

### Exercise 5.29: Crime and punishment

Use `crime`, a panel of 90 North Carolina counties for 1981-1987, and select **1986 only**. Estimate a model relating $LCRMRTE$, the log crime rate, to:

- $PRBARR$: arrests divided by offenses.
- $PRBCONV$: convictions divided by arrests.
- $PRBPRIS$: prison sentences divided by convictions.
- $POLPC$: police per capita.
- $WCON$: weekly construction wages.

Write a short report addressing:

- Your expected effect of each regressor on crime.
- The fitted coefficients, their signs and statistical significance.
- Which variables appear most important for deterrence.
- How you might explain the sign of the police coefficient.

### Hints for Exercise 5.29

Select the year carefully: it is coded as `86`, not `1986`.

```
crime_529 <- crime |> filter(year == 86)
model_529 <- lm(
  lcrmrte ~ prbarr + prbconv + prbpris + polpc + wcon,
  data = crime_529
)
summary(model_529)
```

Before looking at the estimates, write down a reasoned expected sign for each coefficient. Arrest, conviction and imprisonment probabilities are commonly linked to deterrence. Police staffing and wages may also capture county characteristics or reverse causality, so their signs need not have a simple causal interpretation.

Because the dependent variable is logged, a change $\Delta x_j$ in a level regressor has the fitted exact percentage association

$$100\left[\exp(b_j \Delta x_j) - 1\right].$$

Choose meaningful changes rather than comparing raw coefficient sizes—for example, 0.10 for a probability, 0.001 for police per capita, or 100 dollars for weekly wages. Variables measured on different scales cannot be ranked by coefficient magnitude alone.

Structure the short report around four points:

1. expected signs and economic reasoning;
2. estimated signs, p-values and model fit;
3. which punishment variables show statistically significant negative associations; and
4. why a positive police coefficient could arise if high-crime counties hire more police or if omitted urban characteristics affect both variables.

Describe the results as conditional associations. This cross-county OLS model does not by itself identify causal policy effects.

### Exercise 5.30: Could Andy's advertising level be optimal?

For Big Andy's Burger Barn, use `andy` and the quadratic model $SALES = \beta_1 + \beta_2 PRICE + \beta_3 ADVERT + \beta_4 ADVERT^2 + e$. Sales and advertising expenditure are in thousands of dollars.

The book's advertising optimum satisfies $\beta_3 + 2\beta_4 ADVERT_0 = 1$.

At 5%, test whether each candidate could be optimal:

**(a)** $ADVERT_0 = 1.75$.

**(b)** $ADVERT_0 = 1.90$.

**(c)** $ADVERT_0 = 2.30$.

Report a p-value for each test.

### Hints for Exercise 5.30

Estimate the quadratic model first:

```
model_530 <- lm(sales ~ price + advert + I(advert^2), data = andy)
```

For a candidate advertising level $a$, the restriction is

$$H_0 : \beta_3 + 2a\beta_4 = 1, \qquad H_1 : \beta_3 + 2a\beta_4 \neq 1.$$

The estimated marginal revenue is $b_3 + 2ab_4$. Its variance is

$$\widehat{\text{var}}(b_3 + 2ab_4) = \widehat{\text{var}}(b_3) + 4a^2\widehat{\text{var}}(b_4) + 4a\widehat{\text{cov}}(b_3, b_4).$$

You can let `deltaMethod()` perform this calculation:

```
library(car)
candidate_530 <- function(a) {
  result <- deltaMethod(
    model_530,
    "b3 + 2*b4*a",
    parameterNames = c("b1", "b2", "b3", "b4"),
    constants = c(a = a)
  )
  estimate <- result[1, "Estimate"]
  se <- result[1, "SE"]
  t <- (estimate - 1) / se
  p <- 2 * pt(-abs(t), df = df.residual(model_530))
  c(estimate = estimate, se = se, t = t, p = p)
}
```

Run the function for 1.75, 1.90 and 2.30. Reject a candidate at 5% if its two-sided p-value is below 0.05. Failure to reject means that the candidate is compatible with the data; it does not prove that it is the true optimum. Also check that the fitted squared-advertising coefficient gives the concavity required for a maximum.

### Exercise 5.31: Bill's journey to work

The `commute5` data describe 249 working days in 2015. Bill leaves Carnegie between 6:30 and 8:00 AM to drive to the University of Melbourne. $TIME$ is travel time in minutes, $DEPART$ minutes after 6:30 AM, $REDS$ red lights encountered, and $TRAINS$ trains waited for.

**(a)** Estimate $TIME = \beta_1 + \beta_2 DEPART + \beta_3 REDS + \beta_4 TRAINS + e$. Report and interpret every coefficient, including the intercept.

**(b)** Find 95% confidence intervals for all coefficients. Are the estimates precise?

**(c)** At 5%, test that expected delay per red light is at least 2 minutes against less than 2 minutes.

**(d)** At 10%, test that expected delay per train equals 3 minutes against a two-sided alternative.

**(e)** Holding other variables fixed, test at 5% that leaving at 7:30 rather than 7:00 AM adds at least 10 minutes to expected travel time against an increase of less than 10 minutes.

**(f)** At 5%, test that expected delay per train is at least three times the delay per red light against a smaller multiple.

**(g)** With six red lights and one train, test at 5% that a 7:00 AM departure gets Bill to the university by 7:45 AM in expected-time terms, against the alternative that it does not.

**(h)** If being late is unacceptable, are the hypotheses in (g) the appropriate way round? What happens if they are reversed?

### Hints for Exercise 5.31

Estimate the full model with:

```
model_531 <- lm(time ~ depart + reds + trains, data = commute5)
summary(model_531)
confint(model_531)
```

**(a)** `depart` is minutes after 6:30 AM. Thus $b_2$ is the fitted increase in travel time from leaving one minute later, holding lights and trains fixed. Interpret $b_3$ and $b_4$ per additional light and train. The intercept is the fitted travel time for a 6:30 departure with zero lights and zero trains.

**(b)** Use the interval widths as well as whether they include zero when discussing precision.

For **(c)**–**(h)**, a useful general statistic for a linear combination $L = c'\beta$ is

$$t = \frac{c'b - r}{\sqrt{c'\widehat{\mathrm{Var}}(b)c}},$$

where $r$ is the null value. `car::deltaMethod()` can calculate the estimate and standard error of each combination.

**(c)** Test $H_0 : \beta_3 \geq 2$ against $H_1 : \beta_3 < 2$. Use a lower-tail p-value evaluated at the boundary $\beta_3 = 2$.

**(d)** Test $H_0 : \beta_4 = 3$ against $H_1 : \beta_4 \neq 3$. This is a two-sided 10% test.

**(e)** A 7:00 departure has `depart = 30`; a 7:30 departure has `depart = 60`. The fitted difference is therefore $30\beta_2$. Test $H_0 : 30\beta_2 \geq 10$ against $H_1 : 30\beta_2 < 10$.

**(f)** "A train is at least three times a red light" gives $H_0 : \beta_4 - 3\beta_3 \geq 0$ against $H_1 : \beta_4 - 3\beta_3 < 0$. Include the covariance between the two estimates when calculating the standard error.

**(g)** With `depart = 30`, six lights and one train, expected travel time is

$$m = \beta_1 + 30\beta_2 + 6\beta_3 + \beta_4.$$

A 7:00 departure leaves 45 minutes until 7:45. If "does not get there by 7:45" is the alternative, test $H_0 : m \leq 45$ against $H_1 : m > 45$.

**(h)** When lateness is unacceptable, it may be more sensible to require evidence that $m < 45$: test $H_0 : m \geq 45$ against $H_1 : m < 45$. Explain why the decision can differ after reversing the hypotheses. Finally, distinguish an expected-time test from an individual-journey prediction interval:

```
predict(
  model_531,
  newdata = tibble(depart = 30, reds = 6, trains = 1),
  interval = "prediction"
)
```

### Exercise 5.32: Partial out lights and trains

Use `commute5` and the model from Exercise 5.31: $TIME = \beta_1 + \beta_2 DEPART + \beta_3 REDS + \beta_4 TRAINS + e$. Bill is mainly interested in $\beta_2$: he controls departure time, but not the red lights or trains.

**(a)** Regress $DEPART$ on $REDS$ and $TRAINS$, including an intercept, and save the residuals. Which coefficients differ from zero at 5%? Do any significant relationships have a causal interpretation?

**(b)** Regress $TIME$ on $REDS$ and $TRAINS$, including an intercept, and save the residuals. Compare the light and train coefficients with those in the full model that includes departure time.

**(c)** Use the two sets of residuals to estimate $\beta_2$. Adjust the regression output to obtain its correct standard error.

### Hints for Exercise 5.32

This exercise demonstrates the Frisch–Waugh–Lovell result: remove the linear contribution of the control variables from both the outcome and the regressor of interest, then relate the remaining parts.

**(a)** Run the auxiliary regression and save its residuals:

```
depart_532 <- lm(depart ~ reds + trains, data = commute5)
depart_res_532 <- resid(depart_532)
summary(depart_532)
```

Use two-sided 5% tests for the coefficient assessment. Interpret significant terms as associations only: the timing of trains and departures may both be related to the timetable or traffic conditions.

**(b)** Repeat the procedure for travel time:

```
time_532 <- lm(time ~ reds + trains, data = commute5)
time_res_532 <- resid(time_532)
summary(time_532)
```

Compare the `reds` and `trains` coefficients with those from `lm(time ~ depart + reds + trains, data = commute5)`. Think about which part of the departure-time association is absorbed by the reduced model.

**(c)** Regress the two residual series without an intercept:

```
partial_532 <- lm(time_res_532 ~ depart_res_532 - 1)
coef(partial_532)
```

The slope should match the `depart` coefficient in the full model. The raw residual-regression standard error uses the wrong residual degrees of freedom. The full model estimated four parameters, so use $n - 4$:

$$se(b_2) = \sqrt{\frac{\sum_i \hat{e}_i^2}{(n-4)\sum_i \widetilde{DEPART}_i^2}}.$$

Here $\widetilde{DEPART}$ denotes the residuals from part (a), and $\hat{e}$ the residuals from the no-intercept regression in part (c). Verify that the corrected standard error matches the one reported by the full model.

### Exercise 5.33: Education, experience and log wages

Using `cps5_small`, estimate $\ln(WAGE) = \beta_1 + \beta_2 EDUC + \beta_3 EDUC^2 + \beta_4 EXPER + \beta_5 EXPER^2 + \beta_6(EDUC \times EXPER) + e$. Education and experience are in years.

**(a)** At what significance levels do the coefficients differ from zero?

**(b)** Derive the marginal effect of education on expected log wage. How does its estimate change with education and experience?

**(c)** Calculate that effect for every sample observation and draw a histogram. Discuss; report its median, 5th and 95th percentiles.

**(d)** Derive the marginal effect of experience on expected log wage. How does its estimate change with education and experience?

**(e)** Repeat (c) for the experience effect.

**(f)** David has 17 years of education and 8 years of experience; Svetlana has 16 and 18. At 5%, test that Svetlana's expected log wage is at least David's against the alternative that David's is higher. State the hypotheses in terms of the parameters.

**(g)** Repeat the comparison after both gain eight years of experience and no further education. Is the conclusion unchanged? Explain.

**(h)** Wendy has 12 years of education and 17 of experience; Jill has 16 and 11. At 5%, test equality of their marginal experience effects against a two-sided alternative. State the parameter restrictions.

**(i)** How many additional years of experience must Jill gain before her marginal experience effect becomes negative? Find a 95% confidence interval for this quantity.

### Hints for Exercise 5.33

Estimate the model and store its coefficients:

```
model_533 <- lm(
  log(wage) ~ educ + I(educ^2) + exper + I(exper^2) + I(educ * exper),
  data = cps5_small
)
b_533 <- coef(model_533)
summary(model_533)
```

The coefficient order is intercept, education, education squared, experience, experience squared and the interaction.

**(a)** Use the two-sided p-value for each coefficient. A coefficient differs from zero at the 10%, 5% or 1% level when its p-value is below the respective threshold.

**(b)** Differentiate with respect to education:

$$ME_{EDUC} = \beta_2 + 2\beta_3 EDUC + \beta_6 EXPER.$$

Inspect the signs of $2b_3$ and $b_6$ to explain how the fitted effect changes with education and experience. Multiplying the derivative by 100 gives an approximate percentage effect.

**(c)** Create the effect for every observation, then use a histogram and quantiles:

```
effects_533 <- cps5_small |>
  mutate(
    educ_effect = b_533[2] + 2*b_533[3]*educ + b_533[6]*exper
  )

ggplot(effects_533, aes(x = 100 * educ_effect)) +
  geom_histogram(bins = 25, color = "white")

quantile(effects_533$educ_effect, c(0.05, 0.50, 0.95))
```

The percentiles describe heterogeneity in fitted effects across workers; they are not a confidence interval for one parameter.

**(d)** Differentiate with respect to experience:

$$ME_{EXPER} = \beta_4 + 2\beta_5 EXPER + \beta_6 EDUC.$$

Use the signs of $2b_5$ and $b_6$ to describe how this effect changes.

**(e)** Add `exper_effect` to the data frame using the formula in (d), then repeat the histogram and quantile commands from (c). Also check how many fitted effects are below zero.

**(f)** Define $D$ as David's expected log wage minus Svetlana's. Substituting their characteristics and simplifying gives

$$D = \beta_2 + 33\beta_3 - 10\beta_4 - 260\beta_5 - 152\beta_6.$$

The requested hypotheses are $H_0 : D \leq 0$ against $H_1 : D > 0$. Use `deltaMethod()` for the estimate and standard error, then calculate an upper-tail p-value.

**(g)** After eight more years, the same difference becomes

$$D_8 = \beta_2 + 33\beta_3 - 10\beta_4 - 420\beta_5 - 144\beta_6.$$

Test $H_0 : D_8 \leq 0$ against $H_1 : D_8 > 0$. Equal additions to experience do not cancel because the model contains both squared experience and an education–experience interaction.

**(h)** Wendy's marginal experience effect minus Jill's simplifies to

$$12\beta_5 - 4\beta_6.$$

Use a two-sided test of whether this expression equals zero.

**(i)** Hold Jill's education at 16 and solve her experience derivative equal to zero. Her additional years to the turning point are

$$A = -\frac{\beta_4 + 16\beta_6}{2\beta_5} - 11.$$

Use `deltaMethod(model_533, "-(b4 + 16*b6)/(2*b5) - 11", ...)` to obtain the estimate and its delta-method standard error. Form the approximate 95% interval using the residual degrees of freedom. Check the sign of $b_5$ to determine on which side of the threshold the marginal effect is negative.